

Saman Khazaei

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Google Scholar 
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Education

- **Doctor of Philosophy in Biomedical Engineering** New York, USA
New York University (NYU) - Advisor: Dr. Rose T. Faghih *Jan 2022 - May 2026 (expected)*
- **Master of Science in Electrical Engineering** Texas, USA
University of Houston - Advisor: Dr. Rose T. Faghih *Sep 2019 - Dec 2021*
- **Bachelor of Science in Mechanical Engineering** Tehran, Iran
Sharif University of Technology *Sep 2014 - Jul 2019*

Academic Appointments

- **Graduate Research Assistant** New York, USA
New York University. NYU Langone Health - Full time *Jan 2022 - Present*
- **Graduate Research Assistant** Texas, USA
University of Houston - Full time *Sep 2021 - Dec 2021*

Industry Experience

- **Pharmacokinetics/Pharmacodynamics Modeling Intern** Massachusetts, USA
Takeda Pharmaceutical Company (Oncology Dept) - Intern *Jun 2025 - Aug 2025*
- **Renewable Energy Intern** Texas, USA
National Oilwell Varco (NOV) Inc. R&D Department - Intern *Jun 2020 - Jul 2021*

Journal Articles

- **Khazaei, S.**, Gong, J., and Faghih, R. T. (2026). Decoder Design for Concurrent Estimation of Arousal and Performance from One Continuous and Two Binary Observations. *IEEE Transactions on Biomedical Engineering*.
- Xiang, Q., **Khazaei, S.**, and Faghih, R. T. (2025). A leptin-based Bayesian inference of a pro-satiety state reflects a basal circadian rhythm in women with obesity. *Frontiers in Endocrinology*, 16, 1638568.
- **Khazaei, S.** and Faghih, R. T. (2024). Decoding a cognitive performance state from behavioral data in the presence of auditory stimuli. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*.
- **Khazaei, S.** and Faghih, R.T., 2024. Eye tracking is more sensitive than skin conductance response in detecting mild environmental stimuli. *PNAS nexus*, 3(9), p.pgae370.
- Alam, S., **Khazaei, S.** and Faghih, R.T., 2024. Unveiling productivity: The interplay of cognitive arousal and expressive typing in remote work. *Plos one*, 19(5), p.e0300786.
- Amin, M.R., Alam, S., **Khazaei, S.**, Azgomi, H.F. and Faghih, R.T., 2024. Skin Conductance Response Artifact Reduction: Leveraging Accelerometer Noise Reference and Deep Breath Detection. *IEEE Access*.
- **Khazaei, S.**, Amin, M.R., Tahir, M. and Faghih, R.T., 2024. Bayesian inference of hidden cognitive performance and arousal states in presence of music. *IEEE Open Journal of Engineering in Medicine and Biology*.
- **Khazaei, S.**, Parshi, S., Alam, S., Amin, M.R. and Faghih, R.T., 2024. A multimodal dataset for investigating working memory in presence of music: a pilot study. *Frontiers in Neuroscience*, 18, p.1406814.
- Wickramasuriya, D.S., **Khazaei, S.**, Kiani, R. and Faghih, R.T., 2023. A Bayesian filtering approach for tracking sympathetic arousal and cortisol-related energy from marked point process and continuous-valued observations. *IEEE Access*, 11, pp.137204-137247.
- Fekri Azgomi, H., F. Branco, L.R., Amin, M.R., **Khazaei, S.** and Faghih, R.T., 2023. Regulation of brain cognitive states through auditory, gustatory, and olfactory stimulation with wearable monitoring. *Scientific reports*, 13(1), p.12399.

Public Repositories

- **Khazaei, S.**, Parshi, S., Alam, S., Amin, M. R., and Faghih, R. T. (2025). A Multimodal Dataset for Investigating Working Memory in Presence of Music (version 1.0.0). PhysioNet.
- Fekri Azgomi, H., Branco, L., Amin, M. R., **Khazaei, S.**, and Faghih, R. T. (2023). Regulation of Brain Cognitive States through Auditory, Gustatory, and Olfactory Stimulation with Wearable Monitoring (version 1.0.0). PhysioNet.

Conference Proceedings

- **Khazaei, S.** and Faghih, R.T., 2024, November. Decoding a Hidden Energy State Based on Marked Point Process Cortisol Secretory Events During Cardiac Surgery. In 2024 IEEE International Conference on E-health Networking, Application & Services (HealthCom) (pp. 1-6). IEEE.
- **Khazaei, S.** and Faghih, R.T., 2024, October. An Adaptive Marked Point Process Filtering Approach for Decoding Cognitive Performance. In 2024 58th Asilomar Conference on Signals, Systems, and Computers (pp. 950-956). IEEE
- Reddy, R., **Khazaei, S.** and Faghih, R.T., 2023, July. A Point-Process Approach for Tracking Valence using a Respiration Belt. In 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 1-7). IEEE.
- **Khazaei, S.**, Amin, M.R. and Faghih, R.T., 2022, October. Decoding a Neurofeedback-Modulated Performance State in Presence of a Time-Varying Process Noise Variance. In 2022 56th Asilomar Conference on Signals, Systems, and Computers (pp. 990-996). IEEE.
- **Khazaei, S.**, Amin, M.R. and Faghih, R.T., 2021, November. Decoding a neurofeedback-modulated cognitive arousal state to investigate performance regulation by the yerkes-dodson law. In 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 6551-6557). IEEE.

Patents

- Faghih, R. and **Khazaei, S.**, New York University NYU, 2025. System and methods for estimating interoceptive awareness state using eye-tracking measurements. U.S. Patent Application 63/584,814.
- Faghih, R., Reddy, R. and **Khazaei, S.**, New York University NYU, 2025. System and method for estimating emotional valence based on measurements of respiration. U.S. Patent Application 18/778,000.

Selected Honors and Awards

- **3rd place in the 2025 NYU Tandon Research Excellence Exhibit** - Areas of Excellence *Apr 2025*
- **1st place in the 2023 NYU Tandon Research Excellence Exhibit** - Areas of Excellence *Apr 2023*
- **Dissertation Fellowship** - Department of Biomedical Engineering at the NYU *Nov 2021*

Mentorship and Teaching Experience

- **Mentor for Neural and Physiological Signal Processing Course** New York University
Topics: modeling, theory and estimation of point processes, maximum likelihood estimation *Fall 2024*
- **Mentor for Biomedical Modeling, Estimation, & Control Course** New York University
Topics: modeling, stability, controllability, observability, feedback control, optimal control *Spring 2024*
- **Mentoring Master and first-year PhD Students** New York University
Mentees accomplished their academic goal (defending MS degree, advancing to PhD candidacy) *Fall 2022-2023*

Presentations and Invited Talks

- **IEEE International Conference on e-health Networking, Applications and Services** *Nov 2024*
- **IEEE Asilomar Conference on Signals, Systems, and Computers** *Oct 2024 and Oct 2022*
- **IEEE OJEMB Webinar on Computational Modeling & Digital Twin Technology** *Jul 2024*
- **IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)** *Oct 2023*
- **International Conference of IEEE Engineering in Medicine & Biology Society (EMBC)** *Nov 2021*

Professional Service

- **Reviewer for Scientific Reports** *Nov 2025*
- **Reviewer for IEEE-EMBS International Conference on Biomedical and Health Informatics** *Oct 2025*
- **Reviewer for IEEE Asilomar Conference on Signals, Systems, and Computers** *Apr 2025 and Jun 2024*
- **Reviewer for IEEE Transactions on Neural Systems & Rehabilitation Engineering Journal** *Apr 2025*